Distance and Displacement

- Distance: The actual length of the path traveled by an object. It is a Scalar quantity (only magnitude).
- Displacement: The minimum distance between the starting and finishing points. It is a Vector quantity (magnitude and direction).

Speed and Velocity

- Speed: The distance covered by an object in unit time.
  - $\text{Speed} = \frac{\text{Total Distance}}{\text{Time Taken}}$
- Velocity: The displacement that occurs in unit time.
  - $\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$

Acceleration

Acceleration is the rate of change of velocity.

$$\text{Acceleration} (a) = \frac{\text{Final Velocity} (v) - \text{Initial Velocity} (u)}{\text{Time} (t)}$$

- Positive Acceleration: Velocity increases with time.
- Negative Acceleration (Retardation): Velocity decreases with time.
- Zero Acceleration: Velocity remains constant.

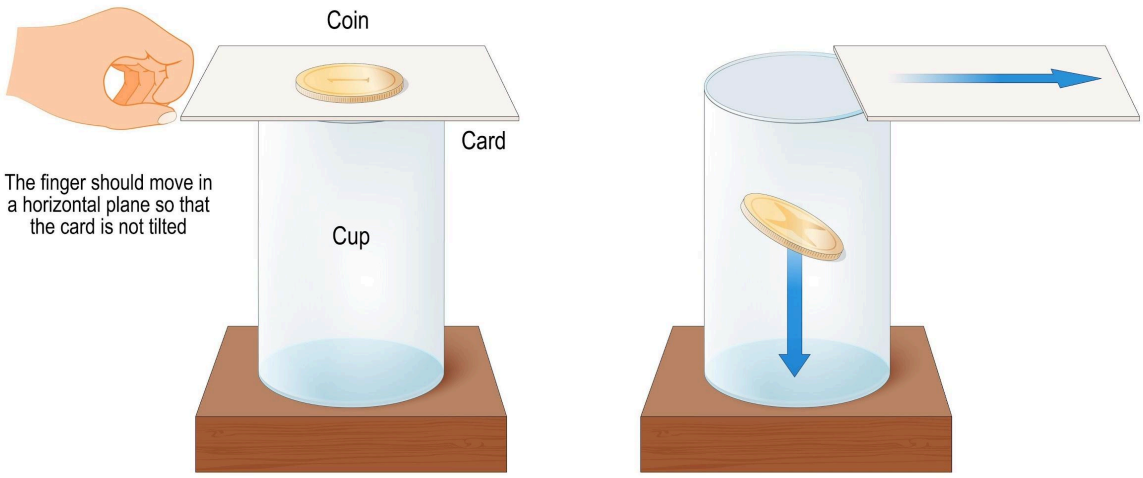
Newton's Laws of Motion

1. Newton's First Law (Law of Inertia)

An object continues to remain at rest or in a state of uniform motion along a straight line unless an external unbalanced force acts upon it.

- Inertia: The inherent property of an object to resist changes in its state of motion.

NEWTON'S FIRST LAW
(experiment)



2. Newton's Second Law (Law of Momentum)

The rate of change of momentum is proportional to the applied force, and the change of momentum occurs in the direction of the force.

- Momentum (P): The product of mass (m) and velocity (v). $P = m \times v$.
- Mathematical Formula: $F = m \times a$ (Force = Mass $\times$ Acceleration).

3. Newton's Third Law (Action and Reaction)

Every action force has an equal and opposite reaction force which acts simultaneously.

- *Example:* When a rocket is launched, the fuel burning downwards (action) pushes the rocket upwards (reaction).

Uniform Circular Motion

- When an object moves with constant speed along a circular path, its motion is called Uniform Circular Motion. Even though the speed is constant, the velocity changes continuously because the direction of motion changes at every point.

Law of Conservation of Momentum

When no external force acts on two interacting objects, their total momentum remains constant.

$$\text{Total initial momentum} = \text{Total final momentum}$$

$$(m_1u_1 + m_2u_2) = (m_1v_1 + m_2v_2)$$

Important Practice Questions

Q1. A person travels 7 meters East and then 7 meters West. Find the distance and displacement.

- Answer:
  - Displacement = 0 meters (since the starting and ending points are the same).
  - Distance = $7 + 7 = 14$ meters.

Q2. Explain why a passenger falls forward when a moving bus suddenly stops.

- Answer: This is due to the Inertia of Motion. When the bus stops, the lower part of the passenger's body comes to rest with the bus, but the upper part tends to remain in motion, causing them to fall forward.

Q3. Calculate the force required to accelerate a 5 kg mass at 2 m/s^2.

- Solution: Using $F = m \times a$:
  $F = 5\text{ kg} \times 2\text{ m/s}^2 = 10\text{ Newtons (N)}$.